

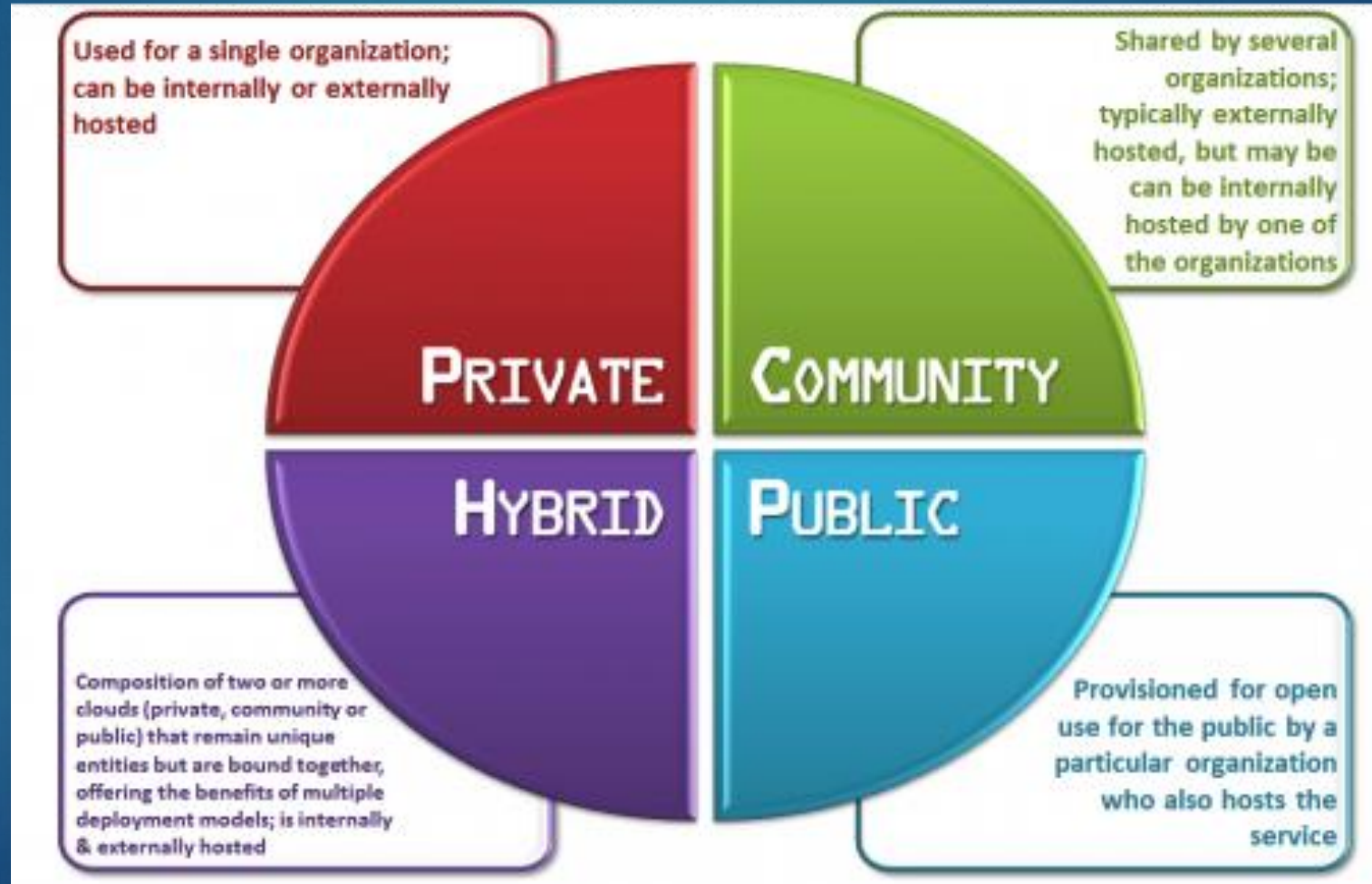


- ▶ You should all have a Slack channel invite
- ▶ Make sure you are using RIT Enterprise Slack
 - ▶ <https://www.rit.edu/its/services/slack>
- ▶ Request to join **RIT-SWEN-614-01-2255**

- ▶ “Cloud computing” is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services).
- ▶ Key principles:
 - ▶ **Pooled resources** - The cloud provider's computing resources are pooled to serve multiple consumers in a multi-tenant environment
 - ▶ **Virtualization** - Each physical server is partitioned into many virtual servers serving multiple customers
 - ▶ **Elasticity** - The ability to dynamically change how much resource is consumed in response to how much is needed
 - ▶ **Automation** - The ability to automatically provision and deploy a new instance of a machine and free or de-provision an instance when not needed
 - ▶ **Metered Billing** - You only pay for resources (servers, disk, etc.) what you use

- ▶ Which type of cloud would best be used by an organization that wants to leverage its existing IT infrastructure but has occasional periods of high demand?

► Cloud Deployment Models



Service Models: SaaS, PaaS & IaaS

Engineering Cloud Software Systems

Department of Software Engineering
Rochester Institute of Technology



Cloud Computing – Who's who?

- ▶ **Cloud providers** are companies that supply the foundational infrastructure, platforms, and enabling technologies for cloud services
 - ▶ They build and maintain the data centers, networks, compute power, and storage needed to support applications, and they offer integration tools for a seamless experience
 - ▶ Examples: AWS, Google Cloud, Microsoft Azure
- ▶ **Service providers** are third-party vendors that develop and deliver specific cloud-based services, applications, or software solutions hosted on cloud providers' infrastructure
 - ▶ They manage and support these services for customers, often offering specialized capabilities or industry-focused applications.
 - ▶ Examples: Microsoft Office 365, Salesforce, Zoom
- ▶ **Customers** are individuals or organizations that use or evaluate services from both cloud providers and service providers
 - ▶ They rely on these services for scalability, cost-effectiveness, and a wide range of applications, from basic infrastructure needs to fully integrated software solutions

Cloud Computing is not a “Silver Bullet”

- ▶ Organizations will need to consider many different factors:
 - ▶ Should I move my existing apps or re-build in the cloud?
 - ▶ Is it more cost effective?
 - ▶ Is it more reliable and resilient?
 - ▶ Is it more secure?
 - ▶ What does my support model look like?
 - ▶ What services should I “Buy” or “Build”?
- ▶ How to properly architect applications to run in the cloud is very critical to ensure success
 - ▶ Doing this **wrong** is easier than you think...



Amazon Prime Day (2018)

SORRY
something went wrong
on our end

Please go back and try again
or go to [Amazon's home page](#).

Hunter
Meet the dogs of Amazon



Seth Fiegerman ✓
@sfiegerman

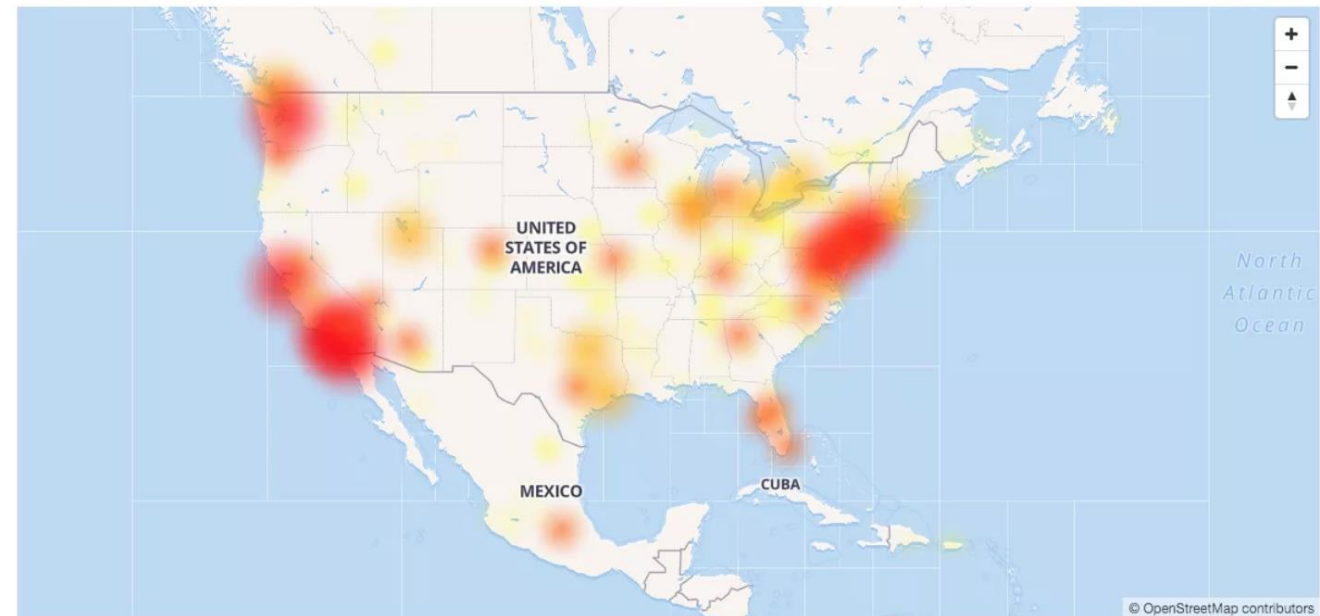


Bad news: Amazon seems to be crashing from demand for Prime Day sales. Good news: Amazon's error page is amazing.

♥ 652 3:07 PM - Jul 16, 2018

Amazon outage map

Amazon outage chart



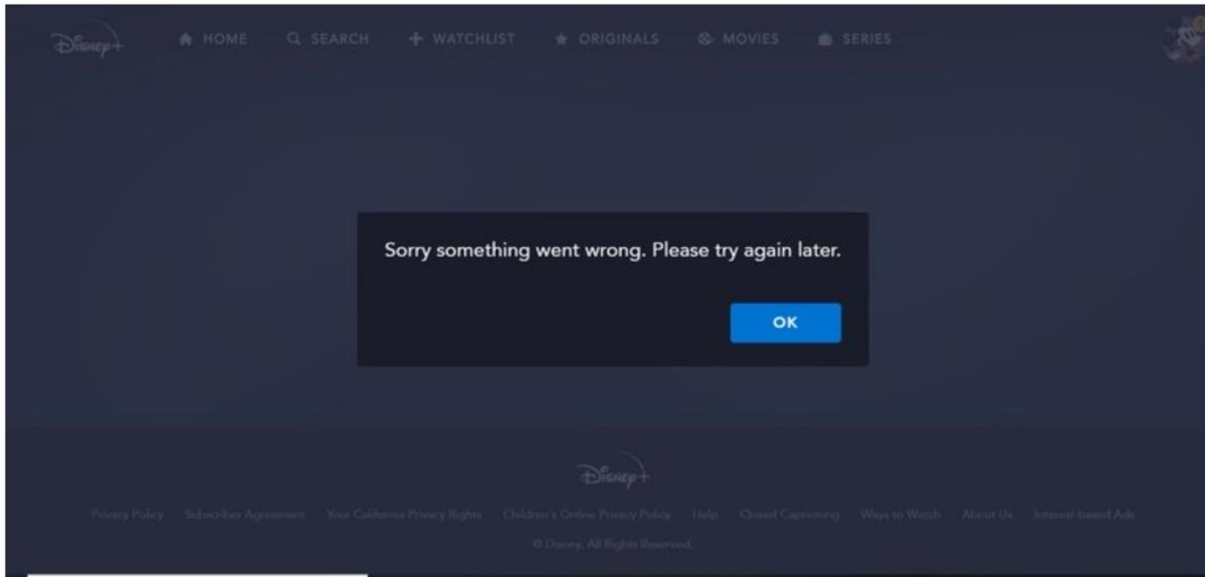
“We now know the reason for Amazon’s brief outage on Prime Day. The retailer reportedly didn’t have enough servers to handle the traffic surge for the day, forcing it to launch a scaled-down version of its homepage and block international traffic, [CNBC reports](#).

According to internal documents, Amazon had to manually add servers to address the issue, something CBNC says may be an indication that the company’s auto-scaling feature failed.”

Disney+ (2019)

9

Disney+ launch day clouded by service availability issues



- ▶ Which cloud provider to blame?
- ▶ Disney+ partners with AWS and Microsoft Azure



Sarah Elizabeth Parker
@ashayamsarah



Honestly not surprised about the [#DisneyPlusfail](#). Even the best servers in the world can't prepare for millions of Americans trying to stream Hannah Montana at once.

11:06 AM · Nov 12, 2019



6.9K 1.3K people are Tweeting about this

But during a service outage, a cloud vendor's role is minimal. As long as the cloud vendor's service is up and operational, "it's the responsibility of the customer to ensure the proper configuration and mechanisms are in place so that services scale appropriately," Gremlin CTO and Co-Founder Matthew Fornaciari

Amazon Says 'Unexpected Behavior' Caused Huge Cloud Outage

- Company's technical statement pledges to learn from event
- Expert says 'bad piece of code' sparked a 'snowball effect'

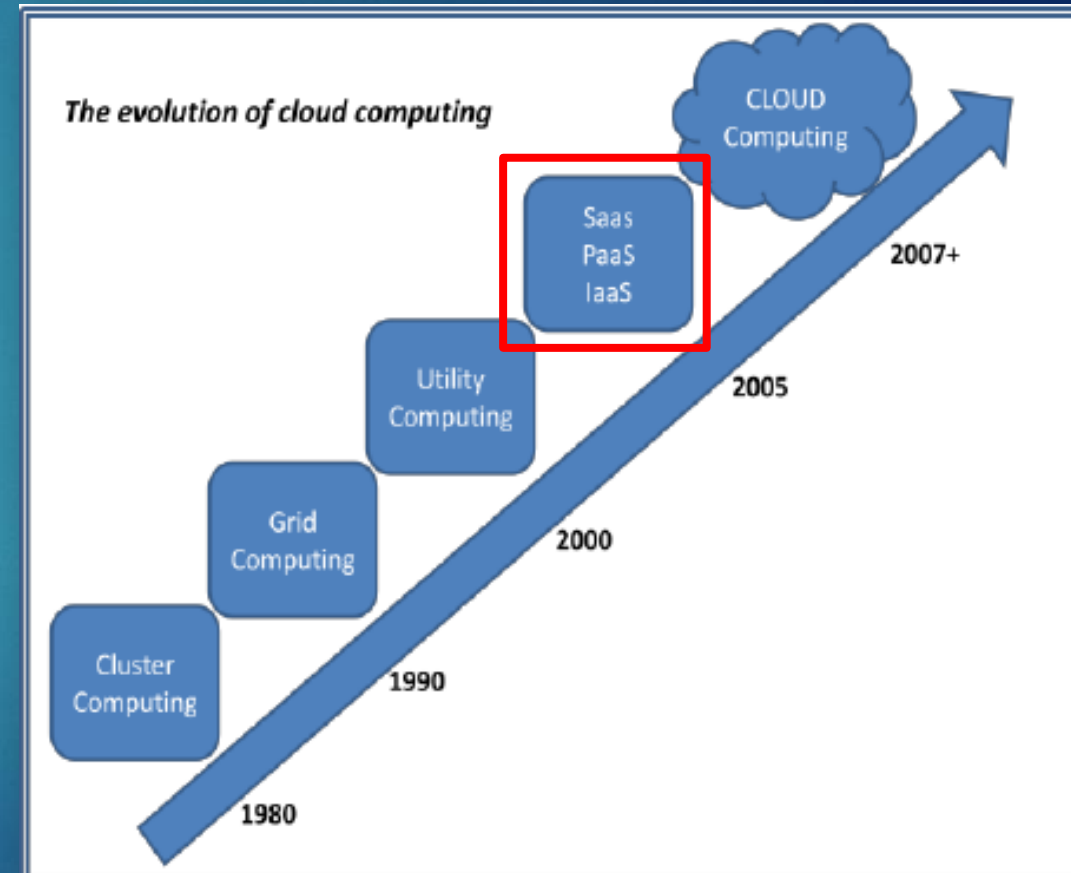
Amazon.com Inc. said automated processes in its cloud computing business caused cascading outages across the internet this week, affecting everything from Disney amusement parks and Netflix videos to robot vacuums and Adele ticket sales.



“Basically, a bad piece of code was executed automatically, and it caused a snowball effect,” Forrester analyst Brent Ellis said. The outage persisted “because their internal controls and monitoring systems were taken offline by the storm of traffic caused by the original problem.”

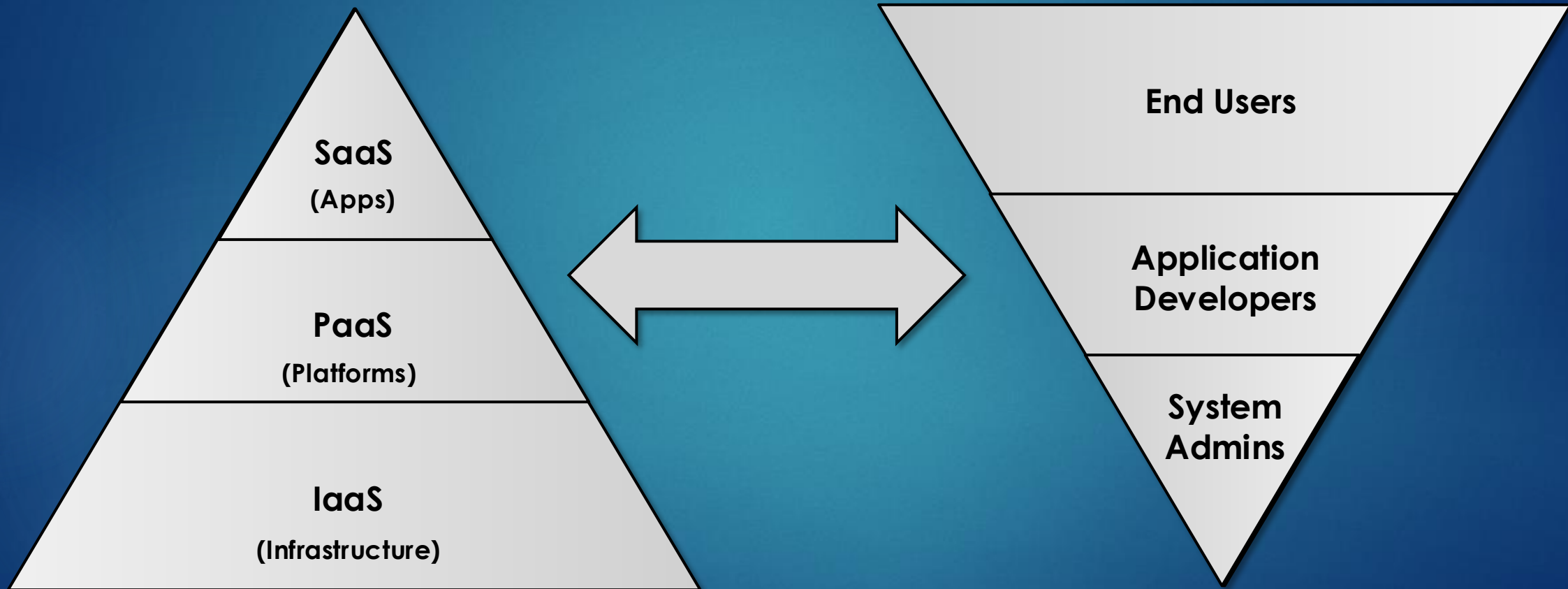
Service Models

- ▶ Cloud providers group their offerings into three primary “As a Service” (aaS) categories according to their level of abstraction:
 1. SaaS - Software as a Service
 2. PaaS - Platform as a Service
 3. IaaS - Infrastructure as a Service
- ▶ These Service Models provide multiple options for organizations to migrate to the cloud faster while minimizing problems



Service Model Pyramid

12



Targeted users

Software as a Service (SaaS)

Software licensing and delivery model in which software is licensed on a subscription basis and is centrally hosted

- ▶ Most mature area of cloud computing
- ▶ SaaS products are generally prebuilt and consumed using the provided functionality without significant customization
- ▶ Many users of SaaS do not realize they are interacting with a cloud at all



SaaS - Example

14

- ▶ Office 365
 - ▶ Subscription-based service
 - ▶ All applications accessible through the Internet on a web browser
 - ▶ Data is stored on OneDrive

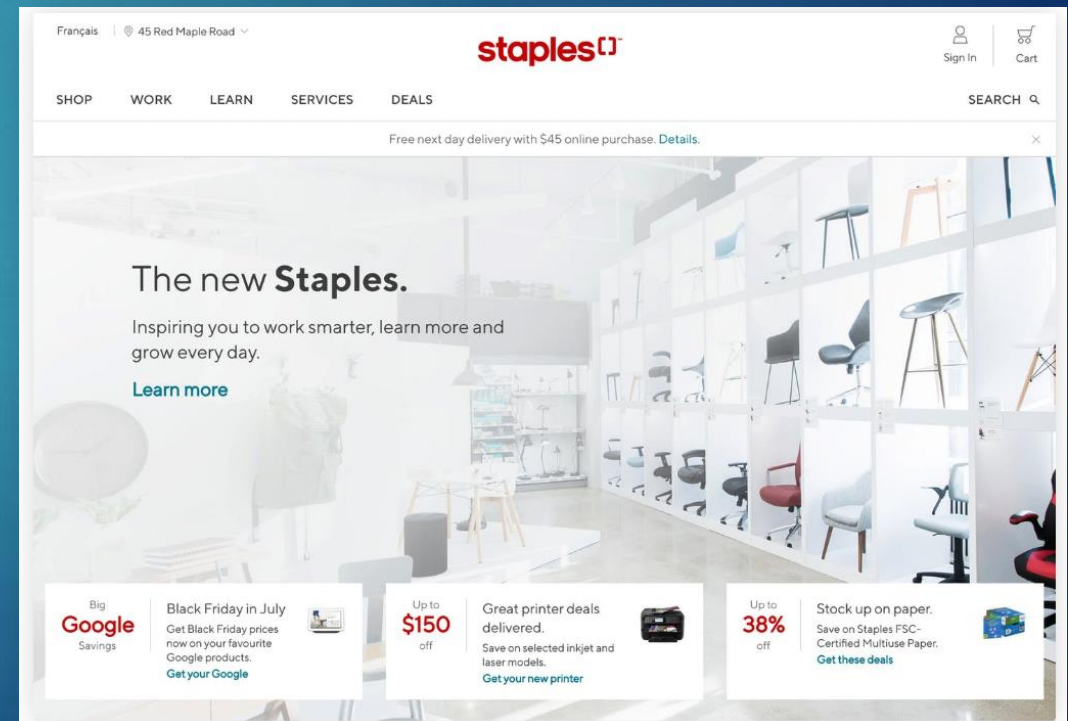


SaaS - Example

15

- ▶ Shopify
 - ▶ Saw the opportunity to create a platform where you could build an ecommerce store and start selling products in just a few hour.
 - ▶ Provides the site, the shopping cart, online catalog, and payment integration all in the one platform

Example: Staples.com built with Shopify

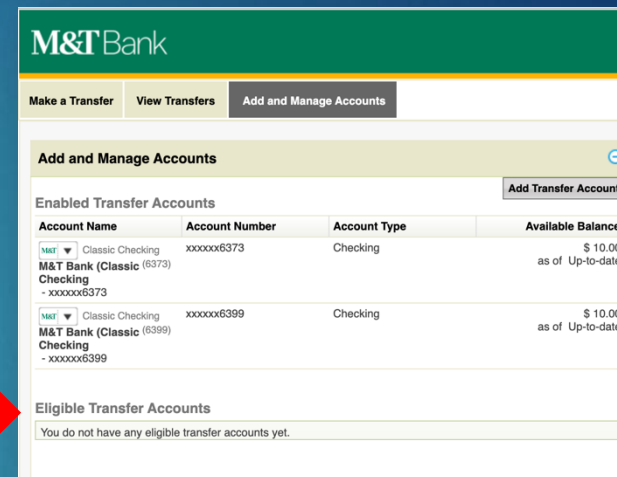
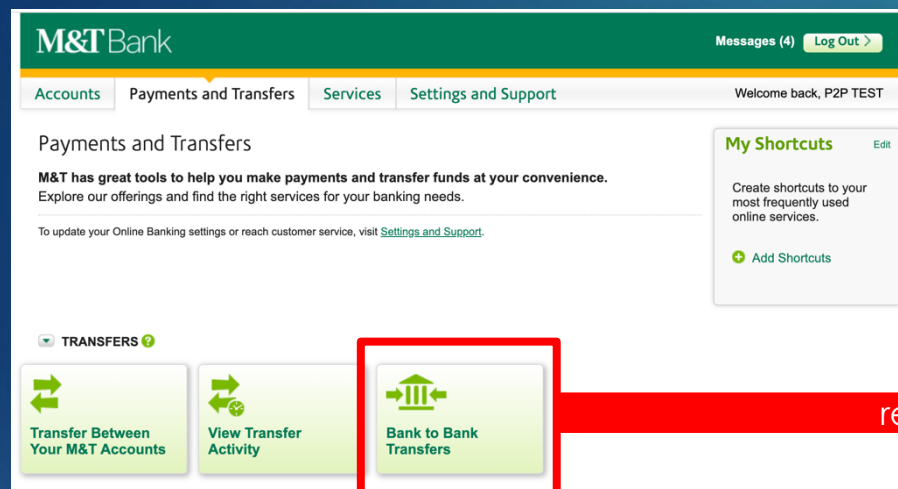


SaaS - Key Characteristics

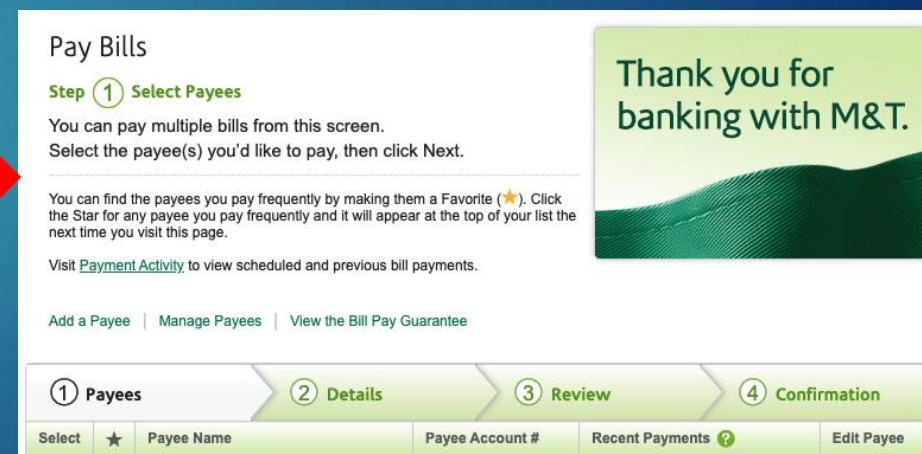
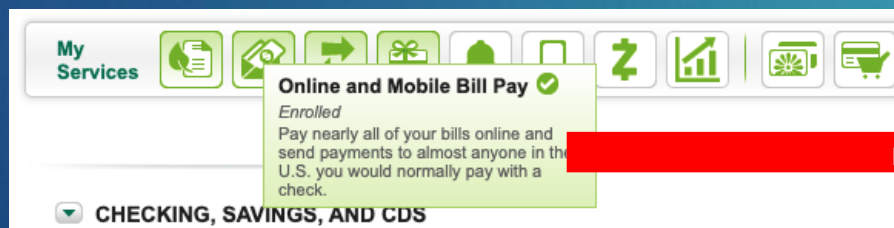
- ▶ Needs to be generalized enough so that lots of customers will be interested in the service
- ▶ Blends sophisticated navigation and ease of use. If isn't easy to use, customers can simply stop subscribing
- ▶ Include measuring and monitoring so customers can be charged actual usage
- ▶ Must ensure that each customer's data and specialized configurations are separate and secure from other customers' data and configurations
- ▶ Need to provide fast releases of new features and new capabilities without impacting the customer's ability to continue business as usual

SaaS Example – M&T Bank (mtb.com)

17



This is managed by **Yodlee**, a SaaS provider that manages bank to bank transfers



BillPay is managed by **FIS**, providing SaaS for Digital Payments

Platform as a Service (PaaS)

A category of cloud computing services that provides a platform allowing customers to develop, run, and manage applications without the complexity of building and maintaining the infrastructure typically associated with developing and launching an app

- ▶ You don't need to directly interact with or administer the virtual OSs
- ▶ Instead, you can let the platform abstract away that interaction and concentrate specifically on writing the application.
- ▶ PaaS systems execute a developer's source code by retrieving all necessary dependencies that are specific to the frameworks and languages used in the application
- ▶ This simplification generally comes at the cost of less flexibility and the requirement to code in the specific languages supported by the particular PaaS provider

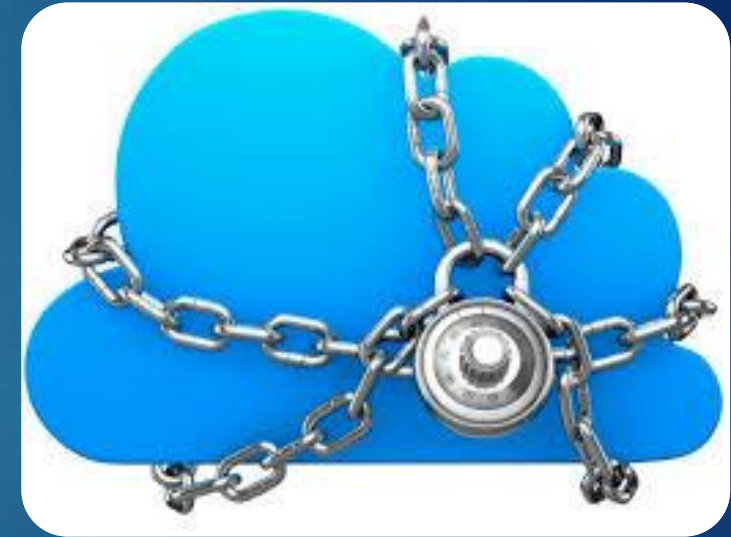


PaaS - Example

- ▶ Google App Engine
- ▶ Google's strategy for PaaS isn't trying to help customers create every application to run on every platform
 - ▶ It's focused on helping customers build Web-based applications
 - ▶ The platform is for development and deployment
- ▶ Other examples: Salesforce.com, Elastic Beanstalk (AWS) and Lightsail (AWS)
- ▶ Several PaaS providers don't charge for development and testing but only when applications are deployed

PaaS – What's not to like?

- ▶ If PaaS providers manage the underlying complexities of the platform and don't charge for developer/testing, what could be an issue?
- ▶ Answer: Vendor lock-in
 - ▶ This makes a customer dependent on a vendor for products and services, unable to use another vendor without substantial switching costs
 - ▶ Vendors can increase costs as they determine
 - ▶ More problematic are new providers that could change or even shut their doors as cloud hosting options evolve with the market



PaaS – continued

- ▶ Because PaaS vendors provide the infrastructure for their application development cloud services, they get to select which languages will be available for application development on their platform
 - ▶ Salesforce.com – Uses a proprietary Apex (Java-like) and Visualforce (XML-like) languages
- ▶ Others in the PaaS space (e.g. Google) are trying to future-proof their offerings by providing support for open languages such as Python and Java to address concerns regarding vendor lock-in



Infrastructure as a Service (IaaS)

Online services that provide high-level APIs used to dereference various low-level details of underlying network infrastructure like physical computing resources, location, data partitioning, scaling, security, backup etc.

- ▶ Allows a customer *almost* complete control over applications, languages, and fundamental resources supporting organizational services such as databases, storage, and networking
- ▶ Customers rent computing resources instead of buying and installing them in their own data center
- ▶ Storage typically charged per gigabyte per month and bandwidth charged for transit into and out of the system
- ▶ An IaaS provider supplies virtual machine images of different operating system flavors



IaaS – What's the appeal?

► Price

- Consider a typical server: rent \$3,600/year OR purchase \$2,000 – \$3,000
- If you purchase, you also need to factor in networking, Internet connections, server management, power, support, etc. (plus pay someone to do it)
- If you rent, this is all covered by the cloud provider

► Security

- Generally, better with cloud providers and can deal with Internet-specific threats such as denial of service (DoS) attack

► Speed to deployment

- Available within minutes!



IaaS - Example

- ▶ Amazon EC2 (Amazon Compute Cloud)
 - ▶ A web-based service that allows businesses to run application programs in the Amazon Web Services (AWS) public cloud
 - ▶ Allows a developer to spin up virtual machines that run with global AWS data centers
 - ▶ Supports several OSs including RedHat Linux, SuSe Linux, Windows Server and even macOS!
- ▶ A simple example...



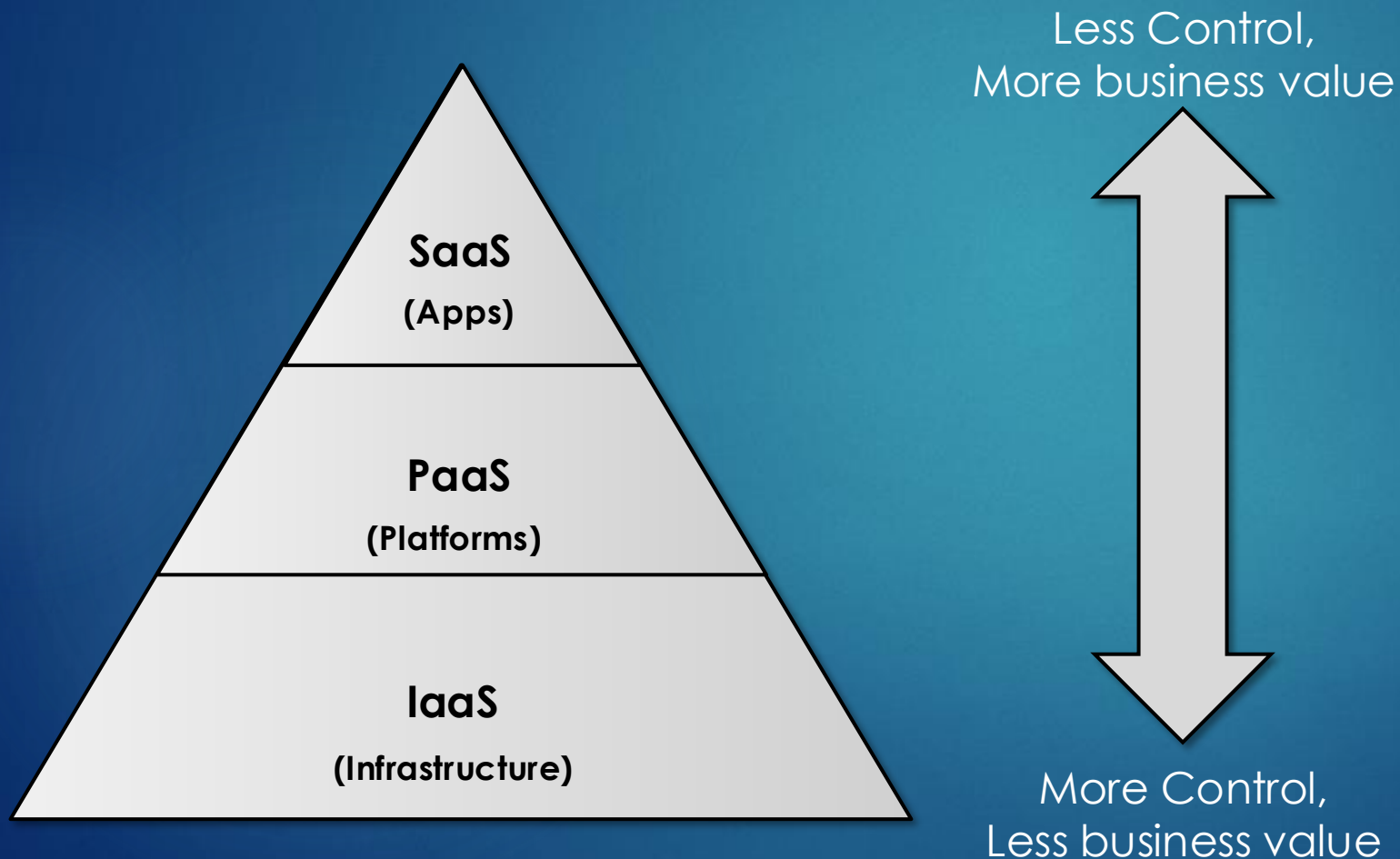
Top 8 IaaS Cloud Provider



Service Models – Control vs. Value

26

- One of the biggest questions organizations will face is “buy vs. build”

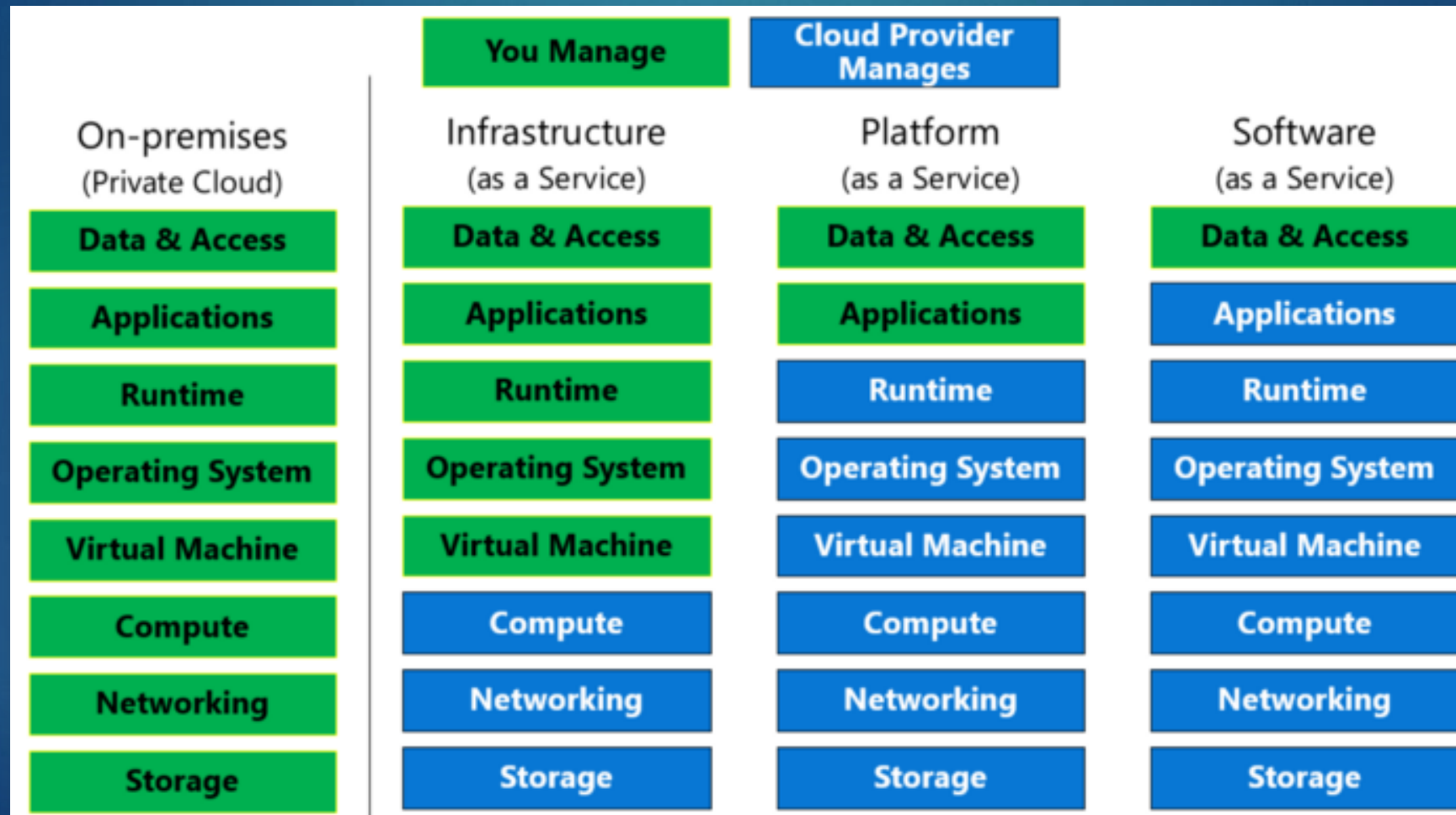


Based on the complexity, organizations may choose one or more service models to build their solution

Summary

27

► On-Premise vs. IaaS/PaaS/SaaS



Summary

28

► Another way to look at SaaS vs. PaaS vs. IaaS





- ▶ The ACME Company, which is a major information provider for News, Financial and Legal content has decided to make a major move to the cloud.
- ▶ They are currently not using any cloud providers.
- ▶ As a cloud consulting group, you have been hired to provide ACME with ideas on how they can achieve this goal:
 - ▶ What specific questions would you ask ACME to help you make your recommendations?
 - ▶ Each team will have 10-15 minutes to discuss a plan

Term Project

- ▶ Teams of 4-5 people
- ▶ Remember to use Slack channel [#project-team-formation](#) to look for a team or if you are a team in need of people
- ▶ Send me Slack or email when you have a team
- ▶ For anyone that cannot find a team by due date, I will randomly assign



- ▶ No class next Monday!
- ▶ Sign up for an AWS “Paid Plan” account
 - ▶ Details on myCourses under Announcements
 - ▶ Note that you will need a credit card to create an account
- ▶ Make sure you have a computer to work on
 - ▶ You just need a browser

Did you create an AWS account before July 15th, 2025? Access the Legacy Free Tier →

AWS Free Tier

Gain free, hands-on experience with AWS products and services

Create a Free Account

New customers get up to \$200 in credits

New AWS customers can get started at no cost with the AWS Free Tier. Gain \$100 USD credits at sign-up and up to \$100 USD more to earn as you explore key AWS services. Test drive AWS services with the Free Plan for up to 6 months. You won't be charged unless you choose the Paid Plan, which allows you to scale your operations and gain access to over 150 AWS services.

Free Plan	Paid Plan
Experience AWS for up to 6 months without cost or commitment	Develop production-ready workloads with access to over 150 AWS services
✓ Receive up to \$200 USD in credits	✓ Receive up to \$200 USD in credits
✓ Includes free usage of select services	✓ Includes free usage of select services
✓ No charges incurred unless you switch to the Paid Plan	✓ Pay beyond credit thresholds
✗ Workloads scale beyond credit thresholds	✓ Workloads scale beyond credit thresholds
✗ Access to all AWS services and features	✓ Access to all AWS services and features